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$$\begin{aligned}\int_1^2 (x^2 + 4)^{\frac{1}{3}} x dx &= \frac{1}{2} \int_1^2 (x^2 + 4)^{\frac{1}{3}} d(x^2 + 4) \\&= \left[\frac{1}{2} \frac{(x^2 + 4)^{\frac{4}{3}}}{\frac{4}{3}} \right]_1^2 = \left[\frac{3(x^2 + 4)^{\frac{4}{3}}}{8} \right]_1^2 \\&= \frac{3(2^2 + 4)^{\frac{4}{3}}}{8} - \frac{3(1 + 4)^{\frac{4}{3}}}{8} = 6 - \frac{3 \cdot 5^{\frac{4}{3}}}{8}\end{aligned}$$

References